

Security risk assessment report

Part 1: Select up to three hardening tools and methods to implement

- **Multifactor Authentication (MFA) :**
Adds an extra layer of security by requiring users to verify their identity using two or more authentication factors.
- **Password policies :**
Enforces the use of strong, unique passwords to minimize the risk of compromised credentials. Regular updates and restrictions against sharing passwords are also critical.
- **Firewall maintenance :**
Ensures the organization's firewall is properly configured to filter unauthorized network traffic and prevent malicious actors from accessing sensitive systems.

Part 2: Explain your recommendations

Enforcing MFA (Multi-Factor Authentication):

Implementing MFA will make our system significantly safer by adding an extra layer of security to user accounts. It reduces the risk of unauthorized access, even if a password is compromised or credentials are stolen. Additionally, it discourages password sharing, as accessing an account would also require a second factor, such as a one-time password (OTP) or biometric authentication, making unauthorized access more burdensome.

Password policies :

Establishing strong password policies strengthens our defenses against brute force and dictionary attacks, particularly if default or weak passwords are being used. Key measures include:

- Enforcing stricter password combinations (e.g., length, complexity).
- Locking accounts after multiple failed login attempts to deter automated

attacks.

- Requiring frequent password updates and prohibiting the reuse of old passwords.

These practices ensure that credentials remain secure and resilient against common attack methods.

Firewall maintenance :

A properly maintained firewall is essential for blocking unauthorized access and suspicious activity within the network. In this case, since the firewall lacks appropriate configuration, it is critical to:

- Establish and enforce rules that block unauthorized inbound and outbound traffic.
- Regularly review and update these rules to adapt to emerging threats.
- Monitor network traffic and logs to identify unusual patterns or potential breaches.

Proper firewall maintenance ensures the network is effectively protected and minimizes the risk of malicious activity.